CLOUD COMPUTING

Module-wise Important Topics & Exam Weightage

Prepared from your uploaded module notes and the October 2025 Cloud Computing previous-year question paper.

Answer-length target supplied by you

- **4-mark answer:** prepare enough material for about 2 pages in your small 19 × 15.5 cm book.
- **8-mark answer:** prepare enough material for about 3–4 pages in the same book.

How to read the priorities

HIGH = directly supported by the previous paper and/or a major topic in the notes. **MEDIUM** = important supporting topic likely to form a short question or part of a long answer. The labels are study priorities, not guaranteed exam predictions.

Important exam pattern seen in the uploaded paper: Part A has five 4-mark questions, one from each module. Part B has 8-mark questions; students answer five full questions, with at least one from each module. Prepare both short and long versions of every HIGH-priority topic.

1. What the previous-year paper tells us

The Cloud Computing paper visible in your uploaded previous-year paper is the October 2025 paper. Its short-answer questions include cloud deployment models, symmetric vs asymmetric clustering, Nova compute components, the ML2 plugin, and Heat. The long-answer section includes logical architecture, cloud service models, OpenStack API services, Horizon, and storage types. The continuation page includes Keystone identity, Nova conductor/scheduler, Swift, Manila, networking, high availability, and orchestration.

Observed 4-mark topics	Observed 8-mark topics
Public, private and hybrid cloud models	OpenStack logical architecture and design considerations
Symmetric vs asymmetric clustering	IaaS, PaaS and SaaS comparison
Nova compute service components	OpenStack API services and how components interact
ML2 plugin in virtual networking	Horizon dashboard and resource management
Heat service and components	Block, object and file-share storage comparison
—	Keystone identity; Nova conductor vs scheduler; Swift ring; Manila; VLAN/tunnel networks; HA; floating IP; Neutron; Heat vs Terraform

Strategy: learn the five 4-mark topics above first because they are directly visible in the paper. Then prepare the 8-mark topics in the module tables below. Do not study only the questions that appeared once; the paper is a guide to emphasis, not a guarantee.

Suggested order if time is limited

Order	What to do
1	Prepare one 4-mark answer for each module (the paper's Part A pattern is one question per module).
2	Prepare at least two 8-mark answers per module, beginning with HIGH-priority topics.
3	Practice diagrams for logical architecture, Neutron architecture, Swift, Magnum and VM booting where included in your notes.
4	Revise definitions, component roles, advantages, working steps, and comparisons for quick recall.

2. Module 1 — Overview of OpenStack

Main coverage in the uploaded notes: introduction to cloud computing, cloud advantages, public/private/hybrid deployment models, IaaS/PaaS/SaaS, OpenStack logical architecture, and core services.

Priority	Topic to cover	Prepare as 4 marks	Prepare as 8 marks
HIGH	OpenStack logical architecture	Define logical architecture; REST APIs; message queue; main services and roles.	Draw and explain the architecture; show how services cooperate to launch/manage cloud resources.
HIGH	Cloud service models: IaaS, PaaS, SaaS	Define each with responsibility and example.	Compare all three, explain responsibility/flexibility/maintenance and how they serve different users.
HIGH	Cloud deployment models: public, private, hybrid	Define and give one real-world example for each.	Compare ownership, access, location, control and flexibility; include examples.
HIGH	OpenStack services: Keystone, Nova, Glance, Neutron, Horizon	Purpose and 3–4 functions of any service.	Explain the major components and how they interact in the logical architecture.
MEDIUM	Message queue and database	Purpose and key functions.	Explain how components communicate and how persistent cloud state is stored.
MEDIUM	Swift, Cinder and Manila	Differentiate object, block and file-share storage.	Compare purpose, characteristics and suitable use cases of each storage service.
MEDIUM	Heat orchestration; Ceilometer/Aodh/Gnocchi	Define Heat or telemetry services.	Explain orchestration using templates/stacks or describe telemetry and metering/alarming.

Likely 4-mark practice: Differentiate public, private and hybrid clouds; explain the purpose of Heat; state the roles of major OpenStack services.

Likely 8-mark practice: Discuss OpenStack logical architecture; compare IaaS, PaaS and SaaS; explain Horizon/API interaction; compare block, object and file-share storage.

3. Module 2 — OpenStack Cluster, Cloud Controller & Common Services

Main coverage in the uploaded notes: clustering, vertical/horizontal scaling, symmetric/asymmetric clustering, cloud controller, Keystone providers, federated identity, Fernet tokens, Nova conductor, Nova scheduler, API services, image management, network service and Horizon.

Priority	Topic to cover	Prepare as 4 marks	Prepare as 8 marks
HIGH	Symmetric vs asymmetric clustering	Definition, node activity, failover and one difference.	Compare architecture, workload handling, failure behavior, scalability and use cases.
HIGH	Keystone identity management	Authentication, authorization, token and service catalog.	Explain identity, resource, authorization, token, catalog and policy providers.
HIGH	Federated Keystone / Federated identity	Definition and two advantages.	Explain the authentication flow and advantages of using an external identity provider.
HIGH	Fernet tokens	Definition and benefits compared with larger database-stored tokens.	Explain token contents, smaller size, speed, lack of database storage, and key rotation.
HIGH	Nova-conductor vs Nova-scheduler	State the role of each and distinguish them.	Explain how conductor protects database access and how scheduler filters/weights hosts.
HIGH	Cloud controller and its services	List the main responsibilities.	Explain the controller as gateway/API, HA/load balancing, database, message queue and storage access.
MEDIUM	Nova API services and Glance image management	Purpose of nova-api or Glance.	Trace API request/image retrieval and the role of metadata and storage backend.
MEDIUM	Vertical vs horizontal scaling	Define both and compare.	Explain scaling approaches and their relationship to cluster capacity and availability.
MEDIUM	Horizon dashboard	Purpose and key features.	Explain dashboard panels, API calls and how it helps users and administrators.

Likely 4-mark practice: Compare symmetric and asymmetric clustering; differentiate Nova conductor and Nova scheduler; explain Fernet tokens.

Likely 8-mark practice: Explain Keystone identity management; discuss federated Keystone and Fernet tokens; explain cloud controller services and Nova conductor/scheduler.

4. Module 3 — OpenStack Compute and Storage

Main coverage in the uploaded slides: Nova components and VM booting, hypervisors and virtualization, Docker containers, Magnum, availability zones, host aggregates, Nova cells/regions, workload segregation, overcommitment, ephemeral/persistent storage, Swift and Cinder.

Priority	Topic to cover	Prepare as 4 marks	Prepare as 8 marks
HIGH	Nova compute components	Roles of nova-api, nova-scheduler, nova-compute and nova-conductor.	Explain the components and sequence involved in launching and managing a VM.
HIGH	Nova-conductor vs Nova-scheduler	Key differences in purpose.	Explain database access separation, host filtering/weighting and VM placement.
HIGH	OpenStack storage: Swift, Cinder, Manila	Define each storage type and its purpose.	Compare object, block and file-share storage, architecture, features and use cases.
HIGH	Swift architecture and Swift ring	Purpose of Swift or the ring.	Explain proxy/account/container/object services, replicas, regions/zones and how the ring maps data to storage nodes.
HIGH	OpenStack Magnum project	Container-as-a-Service, COE and core use.	Explain Magnum architecture/working, COE nodes, Nova, Neutron, Cinder and Heat; include components named in the notes.
MEDIUM	Hypervisors and virtualization	Define hypervisor; list types; compare para- and full virtualization.	Discuss hypervisor choice and compare virtualization approaches.
MEDIUM	Availability zones, host aggregates, Nova cells and regions	Define each briefly.	Compare how each supports fault domains, specialized hosts, scale or separate API endpoints.
MEDIUM	Ephemeral vs persistent storage	State key differences and data persistence.	Compare storage alternatives and discuss external shared vs internal non-shared storage.
MEDIUM	CPU/RAM overcommitment and flavors	Define allocation ratio and flavor.	Explain CPU and RAM overcommitment, default ratios in the notes, benefits and considerations.
MEDIUM	Instance booting and Nova scheduling	List the main steps.	Explain authentication, flavor, scheduling, image retrieval, metadata, network and storage allocation.

Likely 4-mark practice: Describe Nova components; define ephemeral and persistent storage; explain Magnum or availability zones.

Likely 8-mark practice: Compare Swift/Cinder/Manila; explain Swift ring; discuss Magnum; explain VM booting and Nova scheduling; compare cells, regions, availability zones and host aggregates.

5. Module 4 — Neutron and OpenStack Networking

The uploaded Module IV notes are focused on Neutron: its architecture, plugins and agents, VLAN/tunnel networks, virtual switches, ML2, subnets, routers, floating IPs, security groups and Firewall as a Service.

Priority	Topic to cover	Prepare as 4 marks	Prepare as 8 marks
HIGH	Neutron architecture and components	Purpose of Neutron; API server, plugins, agents and extensions.	Draw and explain architecture; describe how requests pass through API, plugin and agents.
HIGH	ML2 plugin	Purpose and significance in virtual networks.	Explain modular Layer 2 design, mechanism/type drivers and how it supports network backends, as described in class notes.
HIGH	VLAN-based vs tunnel-based networks	Compare isolation/operation and give examples from notes.	Explain both network types, virtual switches and how tenant traffic is separated.
HIGH	Floating IPs	Definition, purpose and a simple example.	Explain external-to-private VM connectivity, router/NAT mapping and association steps.
HIGH	Virtual routers and external connectivity	Purpose of router, L3 service/plugin and gateway.	Explain connecting subnets, external gateway, outbound connectivity and inbound access.
HIGH	Security groups and Firewall as a Service	Purpose of security groups or FWaaS.	Explain security rules/policies and firewall configuration/operation using the notes.
MEDIUM	Neutron core and service plugins	Differentiate core plugin and service plugin.	Explain network, port, subnet and service functions such as router/firewall/VPN.
MEDIUM	Neutron agents: L2, DHCP, L3, VPN	List agent roles.	Explain how agents apply network configuration on compute/network nodes.
MEDIUM	Subnets and IP addressing	Define network, subnet, port, CIDR, gateway and DHCP.	Explain network/subnet design and IP allocation with a simple diagram.

Likely 4-mark practice: Explain the ML2 plugin; differentiate VLAN-based and tunnel-based networks; define floating IP; list Neutron components.

Likely 8-mark practice: Explain Neutron architecture; describe floating IPs and external connectivity; compare VLAN/tunnel networks; explain security groups/FWaaS.

6. Module 5 — Notes check required

The uploaded file *CC MOD 5.pptx* contains the title “MODULE 5 CLOUD COMPUTING” but no readable topic text on the remaining slides. I cannot responsibly assign module-specific topics or weightage without the actual Module 5 notes/syllabus.

> **What to do:** check whether the correct Module 5 file was uploaded. If Module 5 covers a topic such as orchestration/Heat or another OpenStack service in your official syllabus, send the proper notes or a clear syllabus image and this section can be completed accurately.

Do not treat Module 5 as unimportant: the previous-year paper pattern explicitly indicates one 4-mark question from each module in Part A and at least one question from each module in Part B.

7. Revision checklist

Done	Revision target
■	I can write public/private/hybrid cloud models and IaaS/PaaS/SaaS without looking at notes.
■	I can draw OpenStack logical architecture and explain Keystone, Nova, Glance, Neutron, Horizon and the message queue.
■	I can compare symmetric/asymmetric clustering and explain the cloud controller.
■	I can distinguish Nova conductor from Nova scheduler and explain Keystone/Fernet/federation.
■	I can compare Swift, Cinder and Manila; explain Swift ring and persistent/ephemeral storage.
■	I can explain Magnum, hypervisors, VM booting and Nova resource segregation.
■	I can draw Neutron architecture and explain ML2, VLAN/tunnel networking, routers and floating IPs.
■	I have prepared at least two 8-mark answers from each available module.
■	I have practiced writing a 4-mark answer in about 2 pages and an 8-mark answer in 3–4 pages of my small book.

Final reminder

The priority ranking combines the uploaded notes with the topics visible in the October 2025 Cloud Computing question paper. It is a revision guide, not a guarantee of what will appear in the next exam. For a 4-mark answer, prepare a definition, 4–6 key points and a small diagram/example where relevant. For an 8-mark answer, prepare an introduction, labelled diagram, components/working, advantages or comparison, use case and conclusion—using only the detail supported by your notes.